

Can an LLM Follow the Evidence in a Graph?

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Quantitative Results Comparison

This section compares Method A (Neighborhood baseline) and Method B (RoG-inspired method) across eight test questions from the RoG-WebQSP dataset. An answer is strictly counted as "supported" only if every predicted answer ends a cited path beginning at a linked entity. Method A generated supported answers only for "Bella Abzug" and "Madrid", while Method B strictly supported "Argentina" and "John F. Kennedy". Method B's other generated answers violated the strict path-ending support rule due to unconstrained LLM output generation.

Table 1: Quantitative Results Comparison

Metric	Method A	Method B
Link Hit@1	87.5% (7 / 8)	87.5% (7 / 8)
Link Hit@3	87.5% (7 / 8)	87.5% (7 / 8)
Answers Generated	2 / 8	5 / 8
Supported Answers	2 / 8	2 / 8
Avg. Triples Shown to LLM	50.0	9.38

Failure Analysis

0.1 Entity Linking Mistake (Method A - Baseline)

- **Question:** "who was the president after jfk died"
- **Analysis:** The question uses the common acronym "jfk", but the target node in the graph is formally labeled "John F. Kennedy". A purely character-based string matching algorithm (like 'difflib.SequenceMatcher') cannot recognize that these two strings refer to the identical semantic entity, as they share almost no character overlap. Without an external alias dictionary or a semantic vector representation, the linker completely misses the ground-truth entity.

Both the words "president" and "jfk" appear as exact substrings within the original question, triggering the "exact match" condition and receiving a base score of 1.0. To break ties, the linker's logic adds a tiny score bonus based on string length (favoring longer, ostensibly more specific entities). Because "President" (9 characters) is longer than "JFK" (3 characters), it receives a higher final score (1.009 vs 1.003) and incorrectly claims the top rank. The pure string-based linker fails to distinguish between a generic noun/title ("President") and the actual core entity ("jfk").

0.2 Unsupported Answers due to LLM Hallucination (Method B)

- **Target Entity:** "Madrid"

- **Analysis:** Analysis: For the Madrid query, Method B outputted an answer array containing 22 distinct tourist attractions (e.g., "El Escorial", "Gran Vía", "Puerta del Sol"). However, the retrieval module only grounded and returned a single evidence path: [["Madrid", "travel.travel_destination.tourist_attractions", "Plaza de Cibeles"]]. According to the strict evaluation rule, because 21 out of 22 predicted answers did not end a cited path, this response is completely unsupported. This failure reveals that despite the RoG grounding module filtering paths, the final answering LLM ignored the restricted context and heavily hallucinated using its internal parametric memory.

Edge-Removal Test: Faithful Reasoning

To evaluate if the LLM faithfully follows graph evidence, we conducted an edge-removal test on the single multi-step question Method B supported correctly.

- **Target Question:** "who was the president after jfk died"
- **Removed Bridge Triple:** ['John F. Kennedy', 'government.us_president.vice-president', 'Lyndon B. Johnson']
- **Method A Behavior:** Method A originally "abstained" due to the "President" linking error, and it continued to "abstain" on the modified graph.
- **Method B Behavior:** On the original graph, Method B outputted "Lyndon B. Johnson" utilizing a valid 2-hop evidence path. When the bridge triple was removed, Method B's physical retrieval traversal was severed. Returning an empty evidence path [], Method B correctly changed its status to "abstain". This proves faithful reasoning: when explicitly blocked by the pipeline, the system refuses to invoke pre-trained knowledge, mitigating hallucination.

Paper Note: Reasoning on Graphs (RoG)

The RoG framework tackles LLM hallucinations and knowledge obsolescence by coupling reasoning with Knowledge Graph (KG) structure. Its core "planning-retrieval-reasoning" idea shifts the paradigm from unguided prompting to generating relation paths as faithful plans, retrieving valid reasoning paths from KGs via graph traversal, and conducting faithful reasoning on the retrieved paths.

In this assignment, I implemented RoG's inference-time pipeline: prompting the LLM for relation plans (Planning), traversing the graph from linked entities (Retrieval), retaining exact matching triples (Grounding), and prompting the LLM with surviving paths or abstaining (Answering).

The eight-question comparison revealed that while Method A suffers from substantial sub-graph noise (50.0 triples avg), Method B drastically reduces context size (9.38 triples avg). Furthermore, the edge-removal test demonstrated that plan-grounded retrieval enforces faithful and verifiable graph reasoning.

Conclusion

The experiment confirms that feeding raw 50-triple sub-graphs (Method A) limits reasoning due to noise and Top-1 entity-linking fragility. While Method B's Plan-then-Retrieve architecture drastically lowered context size (9.38) and demonstrated faithful reasoning during edge-removal, the failure analysis showed that LLMs can still hallucinate answers beyond the retrieved paths. Thus, enforcing strict generation constraints in the final answering step remains critical for a robust KGQA pipeline.